

# de Broglie's Matter Waves, Pauli's Exclusion Principle and Nuclear Binding Energy



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# Whatever we will discuss

- de Broglie's matter waves
- de Broglie's wavelength
- Pauli's exclusion principle
- Nuclear binding energy

## de Broglie's matter waves

Like radiation, matter also has dual (*i.e.*, particle-like and wave-like) characteristics. According to de Broglie's concept of matter waves, moving particle of matter like electron, proton, neutron, atoms or molecules, has always got a wave associated with it and the particle is controlled by the wave in a manner similar to that in which a photon is controlled by waves. **While considering its mass, charge and energy, the electron is considered as a particle, but when the path of a beam of electron or the reflection of a beam of electron is considered, electron is treated as though it were a beam of waves.** These waves, associated with each moving particle of matter, are called **matter waves** or **de Broglie's matter waves** and the wavelength of this wave is called **de Broglie's wavelength**.

# de Broglie's wavelength

According to Plank's quantum theory energy of a photon is,

$$E = h\nu \quad \dots \quad \dots \quad \dots \quad (1)$$

Here,  $h$  = Plank's constant,  $\nu$  = frequency of the photon. If the mass of a photon particle is  $m$ , then according to Einstein's mass-energy relation,

$$E = mc^2 \quad \dots \quad \dots \quad \dots \quad (2)$$

Here,  $c$  = velocity of light.

From equations (1) and (2) we get,

$$E = mc^2 = h\nu \quad \therefore m = \frac{h\nu}{c^2}$$

Suppose the momentum of photon =  $p$

$$\therefore p = \text{mass of photon} \times \text{velocity of the photon}$$

$$= mc = \frac{h\nu}{c^2} \times c = \frac{h\nu}{c} = \frac{h\nu}{\nu\lambda} \quad [\text{As } c = \nu\lambda]$$

$$= \frac{h}{\lambda}$$

# de Broglie's wavelength

$$\text{Or, } \lambda = \frac{h}{p} = \frac{h}{mv}$$

Here,  $m$  = mass of the particle

$v$  = velocity of the particle

This is the famous **de-Broglie's equation**, which establish the relation between particle property of momentum and wave property of wavelength.

# Pauli's exclusion principle

Pauli's Exclusion Principle states that **no two electrons in the same atom can have identical values for all four of their quantum numbers.**

This is an example of a general principle which applies not only to electrons but also to other particles of half-integer spin (**fermions**). It does not apply to particles of integer spin (**bosons**).

The Four Quantum Numbers that describe an Electron

| Number                          | Symbol   | Possible Values                       |
|---------------------------------|----------|---------------------------------------|
| Principal Quantum Number        | $n$      | $1, 2, 3, 4, \dots$                   |
| Angular Momentum Quantum Number | $\ell$   | $0, 1, 2, 3, \dots, (n - 1)$          |
| Magnetic Quantum Number         | $m_\ell$ | $-\ell, \dots, -1, 0, 1, \dots, \ell$ |
| Spin Quantum Number             | $m_s$    | $+1/2, -1/2$                          |

## Pauli Exclusion Principle

- Electrons in a single atom occupy discrete levels of energy. No two “energy levels” or “states” in an atom can have the same energy.
- Each energy level can contain at most two electrons -- one with “clockwise spin” and one with “counterclockwise spin”.
- If two or more atoms are brought together, their outer (i.e., valence) energy levels must shift slightly so they will be different from one another.
- If many (e.g.,  $N$ ) atoms are brought together to form a solid, the Pauli Exclusion Principle still requires that only two electrons in the entire solid have the same energy. There will be  $N$  distinct, but only slightly different valence energy levels, forming a **valence band**.

## Pauli Exclusion Principle Applications

Applied to  
atoms

No two **electrons** in an atom can have identical **quantum numbers**. This guides the modeling of the buildup of the **Periodic Table**.

No two **fermions** can exist in identical energy quantum states

Applied to  
solids

No two electrons in a solid can have identical energy states. This leads to the concept of the **Fermi level** in the **band theory** of solids.

Applied to  
stars

**Electron degeneracy** governs the collapse of **stars** to the **white dwarf** stage.

**Neutron degeneracy** governs the further collapse of stars to the **neutron star** stage.



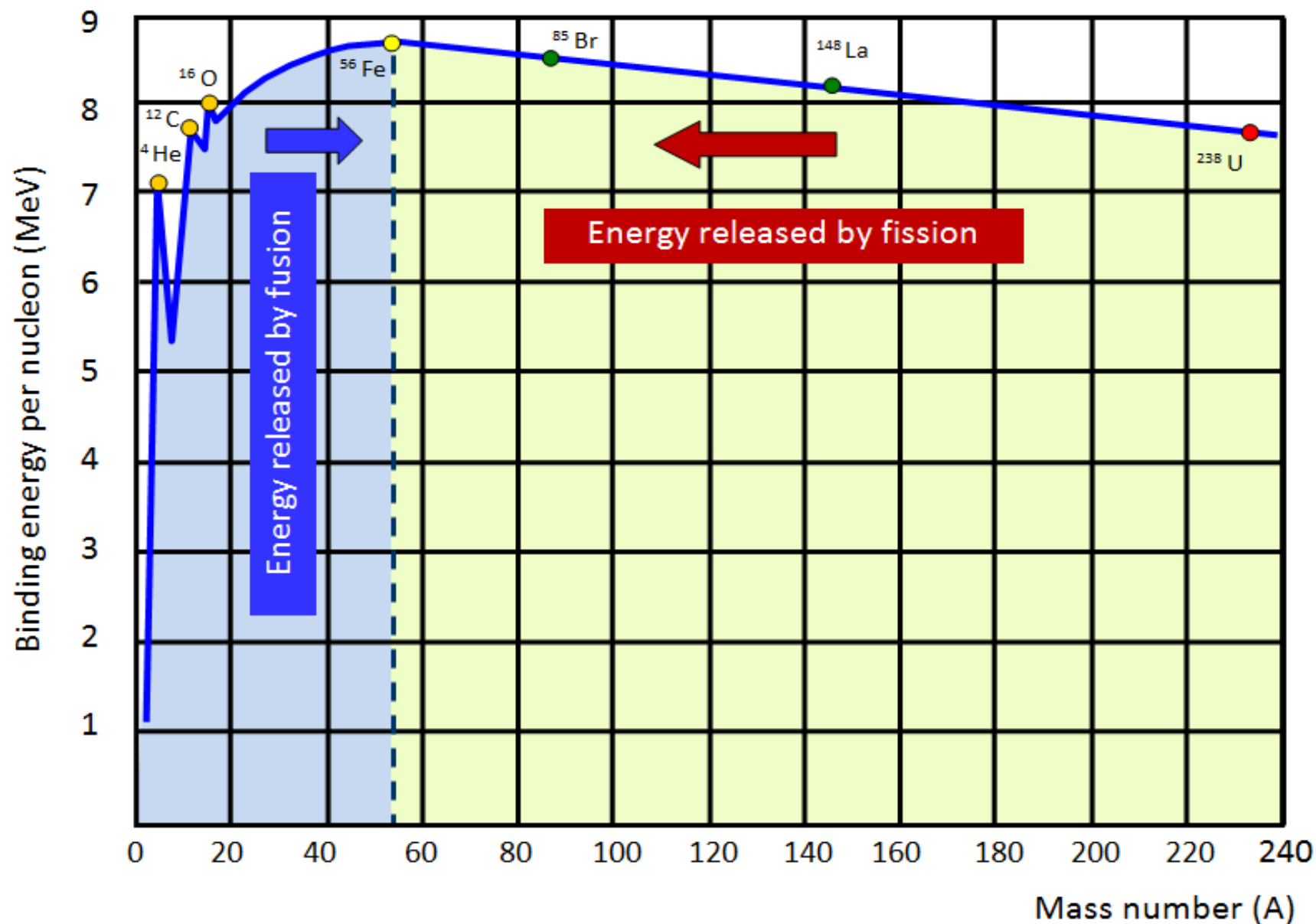
# Nuclear Binding Energy

Measurement of nuclear mass shows that the mass of a nucleus is less than the sum of the masses of the constituent particles. When **Z** protons and **N** neutrons combine to form a nucleus, some of the mass ( $\Delta m$ ) disappears. According to Einstein, this decrease in mass is due to the release of energy when the particles combine to form the nucleus. The energy released is given by the relation  $\Delta E = \Delta m \cdot c^2$ ; where  $\Delta m$  is the decrease in mass and  $c$  is the velocity of light. This energy is called the **Binding Energy** (B.E) of the nucleus.

If **M** is the experimentally determined mass of a nuclide having **Z** protons and **N** neutrons,

$$\text{B.E.} = [(Zm_p + Nm_n) - M]c^2$$

# Binding Energy Graph



# Related Mathematical Problems

de Broglie wavelength related mathematical problems

[Physics for Engineers (Part-2) by Dr. Gias Uddin Ahmad (pp. 630-634)]

**Ex. 15.1, Ex. 15.2, Ex. 15.3 & Ex. 15.5**

Nuclear binding energy related mathematical problems

[Physics for Engineers (Part-2) by Dr. Gias Uddin Ahmad (pp. 783-787)]

**Ex. 18.4, Ex. 18.5 & Ex. 18.6**



**Thank You**  
Thanks for your attention